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Statistical adiabatic channel model for ion-molecule capture processes

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State-resolved and thermally averaged rate coefficients for ion-molecule capture processes are determined by detailed statistical adiabatic channel calculations. The equivalence of this treatment with alternative recent approaches is demonstrated. The advantage of the derived results is their general analytical form which facilitates an application over a wide range of conditions. Particular emphasis is given to ion + nonlinear dipole collisions at low temperatures.

I. INTRODUCTION

There is a strong interest in ion-molecule capture rate coefficients under cool interstellar cloud conditions.¹ Recent laboratory measurements² have arrived at the relevant temperature range below 100 K. A number of theoretical treatments have also appeared.³ It is the aim of the present work to describe analytical expressions for the low temperature limit of ion-molecule capture rate coefficients.⁴

Among the theoretical treatments applied to the problem are ADO theory,^{5,6} variational transition state theory,^{7,8} classical trajectory calculations,⁷⁻⁹ average free energy calculations,¹⁰ scattering calculations using adiabatic capture centrifugal sudden approximations,¹¹⁻¹³ the perturbed rotational state method,¹⁴⁻¹⁹ the adiabatic invariance method,^{6,18,19} and the statistical adiabatic channel model.²⁰ There are close relationships between several of these approaches. The present article uses the statistical adiabatic channel model.²¹ With this general kinetic theory we succeed to derive simple analytical expressions.

In the statistical adiabatic channel model, adiabatic channel potential curves $V_i(r)$ are calculated. Their maxima define the channel threshold energies E_{oi} which, for thermal conditions, lead to the "activated complex" partition function^{20,21}

$$Q^* = \sum_i \exp(-E_{oi}/kT). \quad (1)$$

Using E_{oi} or Q^* , the usual formalism of statistical rate theory (such as transition state theory) is used. For very low temperatures, only the lowest channel states contribute. The potential curves of these channels can be obtained analytically from perturbation theory, such that analytical expressions for channel threshold energies, activated complex partition functions, and capture rate coefficients can be obtained as well.

II. ADIABATIC CHANNEL POTENTIAL CURVES FOR ATOM + LINEAR MOLECULE COLLISIONS

In the adiabatic approximation (for a recent discussion of this approximation in relation to ion-molecule reactions, see Refs. 22 and 23), we assume that the motion in the reaction coordinate is separated from other motions. The distance r between the centers of mass of the colliding species is used as reaction coordinate. The adiabatic channel potential curves $V_i(r)$ are then given by the eigenvalues of the

Schrödinger equation at fixed r . For an atom + linear molecule system this eigenvalue equation in a body-fixed coordinate system²⁴ is written as

$$\left[\frac{B}{\hbar^2} \hat{J}^2 + \frac{|\hat{J} - \hat{j}|^2}{2\mu r^2} + V(r, \vartheta) - V_i(r) \right] \Psi = 0. \quad (2)$$

Here, $\hat{j}$ is the angular momentum operator for rovibrational motion of the linear molecule (rotational constant B in energy units), $\hat{J}$ is the total angular momentum operator, μ is the reduced mass, $V(r, \vartheta)$ the potential with an angle ϑ between the axis of the linear molecule and the line connecting the centers of mass. At $r \rightarrow \infty$, one has $(2j + 1)$ -fold degenerate levels $V_i(r \rightarrow \infty) = B j(j + 1)$. With decreasing r , the m degeneracy is lifted and the channel potential curves split into $V_i(r) = V(r, j, J, m)$. The quantum number m corresponds to the projection of $\hat{j}$ and $\hat{J}$ on the line connecting the centers of mass, i.e., $\hat{J} \cdot \hat{j} = m^2 \hbar^2$. If, e.g., the angular potential for an ion + permanent dipole collision is considered, i.e.,

$$V(r, \vartheta) = -\frac{q\mu_D}{r^2} \cos \vartheta, \quad (3)$$

Eq. (2) can be written as

$$\left[\frac{1}{\sin \vartheta} \frac{d}{d\vartheta} \left(\sin \vartheta \frac{d}{d\vartheta} \right) - \frac{m^2}{\sin^2 \vartheta} \right] \Psi + (\epsilon + \lambda \cos \vartheta) \Psi = 0, \quad (4)$$

where

$$\epsilon = \frac{V_i(r) - l(l + 1)\hbar^2/2\mu r^2}{B}, \quad (5)$$

$$\lambda = \frac{q\mu_D}{Br^2}, \quad (6)$$

and

$$l(l + 1) = j(j + 1) - 2m^2 + J(J + 1). \quad (7)$$

Here q denotes the ionic charge, μ_D is the permanent dipole moment, and l is the quantum number for orbital motion. Solutions of Eq. (2) are well known from the Stark effect of linear dipole molecules. Power series expansions in λ (for large r), in $\lambda^{-1/2}$ (for small r), and tabulations for intermediate r are available.^{25,26} For low temperature applications, we are only considering the lowest channel potential curves. For the limit $T \rightarrow 0$, even only those curves contribute which (for $l = 0$) are purely attractive. These curves are given by second order perturbation theory, e.g., for the ion + permanent dipole case one has

$$\epsilon \approx j(j+1) - \frac{\lambda^2}{2} \left[\frac{3m^2 - j(j+1)}{j(j+1)(2j-1)(2j+3)} \right]$$

$$= j(j+1) - \frac{\lambda^2}{2} F(j, m) \quad (8)$$

or $\epsilon = -\lambda^2/6$ for $j = m = 0$. Only those channels (for $l = 0$) are purely attractive for which $F(j, m)$ is positive.

Equations (5)–(8) lead to the channel potential curves

$$V_i(r) \approx Bj(j+1) + \frac{l(l+1)\hbar^2}{2\mu r^2} - \frac{q^2\mu_D^2}{2Br^4} F(j, m), \quad (9)$$

valid at sufficiently large values of r where second order perturbation theory applies. Complete and accurate adiabatic channel potential curves have been illustrated, e.g., in Refs. 14–17, and 22–26, and are not repeated here. It should be mentioned that only $l = 0$ channels are purely attractive. For $l \neq 0$, channel maxima with $E_{oi} > Bj(j+1)$ always occur.

III. ADIABATIC CHANNEL THRESHOLD ENERGIES E_{oi}

We further consider ion + linear molecule capture processes, but go beyond the charge + permanent dipole case of Eqs. (3)–(9). We include charge + permanent dipole, charge + quadrupole, and charge + induced dipole contributions in the interaction potential by using

$$V(r, \vartheta) = -\frac{q\mu_D}{r^2} \cos \vartheta + \frac{qQ(3\cos^2 \vartheta - 1)}{2r^3}$$

$$- \frac{q^2}{6r^4} [3\alpha + (\alpha_{\parallel} - \alpha_{\perp})(3\cos^2 \vartheta - 1)]. \quad (10)$$

Here Q denotes the quadrupole moment of the linear molecule, α the isotropic polarizability, and $\alpha_{\parallel}$ and $\alpha_{\perp}$ the anisotropic components of the polarizability parallel and perpendicular to the axis of the linear molecule. The corresponding adiabatic channel potential curves from perturbation theory,^{25–28} by analogy with Eq. (9) are given as

$$V_i(r) \approx Bj(j+1) + \frac{l(l+1)\hbar^2}{2\mu r^2} - \frac{q^2\mu_D^2}{2Br^4} F(j, m) - \frac{qQ}{2r^3}$$

$$- \frac{q^2\alpha_{\perp}}{2r^4} + \left[\frac{3qQ}{2r^3} - \frac{(\alpha_{\parallel} - \alpha_{\perp})q^2}{2r^4} \right] \frac{F'(j, m)}{2} \quad (11)$$

with^{26,29}

$$F'(j, m) = 1 - \frac{(2m-1)(2m+1)}{(2j-1)(2j+3)}. \quad (12)$$

The mean polarizability α is defined by $\alpha = (\alpha_{\parallel} + 2\alpha_{\perp})/3$. For isotropic polarizability, i.e., $\alpha = \alpha_{\parallel} = \alpha_{\perp}$ and $\mu_D = 0 = Q$, the channel potential curves have maxima

$$E_{oi} = Bj(j+1) + \frac{[l(l+1)]^2\hbar^4}{8\mu^2\alpha q^2}. \quad (13)$$

Neglecting the quadrupole and polarizability anisotropy contributions, the threshold energies from Eq. (11) follow as

$$E_{oi} \approx Bj(j+1) + \frac{BG^2}{2\{F(j, m) + \alpha B/\mu_D^2\}} \quad (14)$$

with

$$G = \frac{l(l+1)}{l_0(l_0+1)} \quad (15)$$

and

$$l_0(l_0+1) = 2\mu q\mu_D/\hbar^2. \quad (16)$$

Equation (14) is only valid if $F(j, m)$ is positive; in other cases the perturbation treatment does not apply, see below; for $j = m = 0$, $F(j, m)$ is equal to $1/3$. Including all terms of Eq. (11) still leads to analytical general expressions for the channel threshold energies, in as far as they can be expressed by first and second order perturbation theory.

Channel potential curves, for which $F(j, m)$ is negative at large r are repulsive. In this case, not the perturbation treatment but a small r -power expansion around torsional oscillator eigenvalues will be employed. The corresponding series expansions for $\cos \vartheta$ potentials in Eq. (10) are given in Ref. 26, for $\cos^2 \vartheta$ potentials, see Ref. 29. Considering only the permanent dipole part of Eq. (10), the first two terms of the series with Eq. (5) lead to eigenvalues

$$\epsilon \approx -\lambda + a\sqrt{2\lambda} + b/4, \quad (17)$$

where

$$a = 2j - |m| + 1 \quad (18)$$

and

$$b = m^2 + 2|m|j - 2j^2 + |m| - 2j - 2. \quad (19)$$

Equation (17) leads to channel potential curves

$$\frac{V_i(r)}{B} \approx \frac{l(l+1)\hbar^2}{2\mu r^2 B} - \frac{q\mu_D}{r^2 B} + a\sqrt{\frac{2q\mu_D}{r^2 B}} + \frac{b}{4}. \quad (20)$$

The corresponding channel threshold energies are

$$\frac{E_{oi}}{B} \approx \frac{a^2}{2(1-G)} + \frac{b}{4}. \quad (21)$$

For large G , Eq. (21) is also approached for channels whose threshold energies at small G are given by perturbation theory, i.e., by Eq. (14).

IV. LOW TEMPERATURE RATE COEFFICIENTS

With the analytical expressions for channel threshold energies E_{oi} of Eqs. (14) and (21), or the analogous values from tabulations,²⁶ the thermally averaged overall rate coefficients $k(T)$ or state n -resolved rate coefficients $k(n, T)$ are calculated via²⁰

$$k(n, T) = \frac{kT}{h} \left(\frac{h^2}{2\pi\mu kT} \right)^{3/2} Q^*(n), \quad (22)$$

$$Q^*(n) = \sum_{l=0}^{\infty} (2l+1) \exp(-[E_{oi}(n) - E_0(n)]/kT), \quad (23)$$

and

$$k(T) = \frac{\sum_n k(n, T) \exp(-E_0(n)/kT)}{\sum_n \exp(-E_0(n)/kT)}. \quad (24)$$

Here, $E_{oi}(n)$ denotes the state n -resolved threshold energy as a function of the orbital angular momentum quantum number l , and $E_0(n)$ denotes the energy levels of the separated reactants at $r \rightarrow \infty$.

At first we only treat the limit $T \rightarrow 0$. In the presence of a permanent dipole moment μ_D and an isotropic polarizability α , and only treating the $j = m = 0$ channel, with Eqs. (14)–(16) and (22)–(24) integration over l leads to

$$k(T \rightarrow 0) = 2\pi q \sqrt{\alpha/\mu} \sqrt{1 + \mu_D^2/3\alpha B} = k_L \sqrt{1 + y^2/3}, \quad (25)$$

where $y = \mu_D/\sqrt{\alpha B}$. [It should be mentioned that the summation over l leads to a divergence at $T \rightarrow 0$ which, however, is an artifact of the adiabatic treatment (violation of the uncertainty principle for the $j = m = l = 0$ channel.^{27,30})] Likewise, for state n -resolved thermal rate coefficients $k(n, T \rightarrow 0)$, we obtain

$$k(j = |m|, T \rightarrow 0) = k_L \sqrt{1 + F(j, m)y^2}. \quad (26)$$

Equation (26) applies for all channels $|m| \leq j$ where $F(j, m)$ is positive. However, $k(n, T \rightarrow 0) = 0$ is obtained when $F(j, m)$ is negative. Equation (26) also can be derived from the semiclassical adiabatic invariance and PRS methods of Refs. 6, 16, 18, and 19. This confirms that these methods, apart from minor calculational details, are closely related to the statistical adiabatic channel model of Ref. 21, using the same threshold energy behavior of the adiabatic potentials as well as a statistical population of the adiabatic channel states.

After having obtained simple analytical expressions of the thermal rate coefficients in the $T \rightarrow 0$ limit, we extend our treatment to higher temperatures. As before, we express the activated complex pseudopartition function $Q^*(n)$ by an integral. Replacing l by G , see Eqs. (15) and (16), and the l dependence of $E_{0l}(n)$ by a G dependence, i.e., $E_0(n, G)$, we have

$$\begin{aligned} Q^*(n) &= l_0(l_0 + 1) \int_0^\infty \exp\left[-\frac{E_0(n, G) - E_0(n)}{kT}\right] dG \\ &= l_0(l_0 + 1) \exp\left[-\frac{E_0(n, G=0) - E_0(n)}{kT}\right] \\ &\quad \times \int_0^\infty \exp\left[-\frac{E_0(n, G) - E_0(n, G=0)}{kT}\right] dG. \end{aligned} \quad (27)$$

For the attractive channels with $F(j, m) > 0$ and $E_0(n, G=0) = E_0(n)$, the exponential factor in front of the integral equals unity; for the repulsive channels, with $F(j, m) < 0$ and $E_0(n, G=0) > E_0(n)$, it is smaller. Considering only permanent dipole contributions, channel threshold energies for $j \ll l_0$ and $G \rightarrow 1$ approach the results of Eq. (21) such that the upper limit of the integral

$$I = \int_0^\infty \exp\left[-\frac{E_0(n, G) - E_0(n, G=0)}{kT}\right] dG \quad (28)$$

can be put equal to unity. The evaluation of the integral I is straight forward with the perturbation treatment for $T \rightarrow 0$. It gives

$$I \rightarrow I_0 = \sqrt{\frac{\pi kT(1 + y^2 F(j, m))}{2By^2}}. \quad (29)$$

With increasing temperature, the integral I approaches unity. We have evaluated the integral I numerically with accurate channel eigenvalues. The results can well be represented by the simple expression

$$I \approx 1 - \exp\left[-I_0 \left\{\frac{1 + 1.5I_0}{1 + I_0}\right\}\right] \quad (30)$$

which interpolates between $I \approx I_0$ for small I_0 (low temperatures) and unity (high temperatures). This empirical approximation of the integral I reproduces the accurate values satisfactorily well (generally to a few percent) such as demonstrated below.

With increasing importance of induced dipole moments, the result has to be further modified in order to account for the finite barriers at $G > 1$. An approximate treatment of these contributions can be based on the corresponding threshold energies for isotropic potentials,²⁰ i.e., $E_0(G) \approx Bj(j+1) + B(G-1)^2 y^2/2$. Integrating over G from 1 to ∞ in Eq. (28) using this $E_0(G)$, would give an additional term $\sqrt{\pi kT/2By^2}$. However, the anisotropy of the potential leads to a reduction of this contribution. We express this by multiplying the given term by

$$I \exp[-E_0(G=1)/kT]$$

where

$$E_0(G=1) = 3[a(j, m)]^{4/3} y^{2/3} B/4$$

is the threshold energy for the torsional oscillator limit of Eq. (17). Combining Eqs. (14), (16), (22), (25), and (30) with this term, one finally obtains the approximate expression for the state-resolved thermal rate coefficient

$$\begin{aligned} \frac{k(n, T)}{k_L} &\approx \left\{ y \sqrt{\frac{2B}{\pi kT}} + \exp\left(-\frac{3}{4}[a(j, m)]^{4/3} y^{2/3} \frac{B}{kT}\right) \right\} \\ &\quad \times \left\{ 1 - \exp\left[-I_0 \left(\frac{1 + 1.5I_0}{1 + I_0}\right)\right] \right\}, \end{aligned} \quad (31)$$

where $I_0 = \sqrt{\pi kT(1 + y^2 F(j, m))/2By^2}$ from Eq. (29), $k_L = 2\pi q \sqrt{\alpha/\mu}$, $y = \mu_D/\sqrt{\alpha B}$, $a(j, m) = 2j - |m| + 1$, and $F(j, m)$ from Eq. (8) are used as before. For $T \rightarrow 0$, Eq. (31) coincides with Eq. (26).

Equation (31) only applies to channels for which $F(j, m) > 0$, such as all channels with $j = |m|$. For channels with $F(j, m) < 0$ a slightly modified treatment is used. At first, there is a barrier at $G = 0$ such that the exponential factor in front of the integral of Eq. (27) is smaller than unity. Furthermore, with the threshold energies from the torsional oscillator limit of Eqs. (17) and (21) (including only the a term), we interpolate the integral I between I_0 and unity by $I \approx [1 + a^2(j, m)B/2kT]^{-1}$. In addition to this integral between 0 and 1, for the range 1 to ∞ we add a term analogous to Eq. (31). By analogy to Eq. (31), for channels with $F(j, m) < 0$ this leads to

$$\begin{aligned} \frac{k(n, T)}{k_L} &\approx \left\{ y \sqrt{\frac{2B}{\pi kT}} + \exp\left(-\frac{3}{4}[a(j, m)]^{4/3} y^{2/3} \frac{B}{kT}\right) \right\} \\ &\quad \times \exp\left(-\left[\frac{a^2(j, m)}{2} - j(j+1)\right] \frac{B}{kT}\right) \\ &\quad \times [1 + a^2(j, m)B/2kT]^{-1}. \end{aligned} \quad (32)$$

Equations (31) and (32) are good analytical representations of the detailed channel eigenvalue results. They facilitate calculations for situations where many channels contribute. The quality of these equations will be demonstrated

in practical applications below.

Our results for $k(j=0, T)$, $\bar{k}(j=1, T) = 2k(j=1, |m|=1, T)/3 + k(j=1, m=0, T)/3$, and $\bar{k}(j=2, T) = 2\{k(j=2, |m|=2, T) + k(j=2, |m|=1, T)\}/3 + k(j=2, m=0, T)/3$ can be compared with the numerically fitted results from Ref. 19. Over the range $0.1 < kT/B < 10^3$ both treatments nowhere differ from the accurate evaluations by more than 10%. It should be noted that the expressions of Ref. 19 become inappropriate for $kT/B < 0.1$, diverging at $T \rightarrow 0$. This is, however, without practical relevance. Also, analytical expressions in Ref. 19 were fitted only for small j . Expressions for higher j are required in order to calculate overall thermal rate coefficients $k(T)$ for temperatures above $kT/B \gtrsim 10$, e.g., channels up to about $j=20$ markedly contribute^{11,13} to the reaction $\text{H}_3^+ + \text{HCN}$ at 300 K. Our approximate analytical results of Eqs. (31) and (32) can be easily used for high temperature calculations of $k(T)$, see below.

V. EXAMPLES FOR ION + LINEAR DIPOLE COLLISIONS

In the following, we apply our results to the reactions $\text{H}_3^+ + \text{HCN}$ and $\text{H}_3^+ + \text{HCl}$ which have served as test cases before.^{1,13,19,20,31} Figures 1 and 2 show the state-resolved thermal rate coefficients $k(j=0, T)$ and $k(j=1, T)$. The chosen plot as a function of $\sqrt{kT/B}$ magnifies the low temperature range. There is a good agreement between all treatments at $kT/B \gtrsim 20$ confirming their general close relationship. Apart from minor calculational artifacts, for $0.1 \lesssim kT/B \lesssim 20$, the statistical adiabatic channel model and the semi-classical adiabatic invariance method coincide as well. The $T \rightarrow 0$ limits are indicated by asterisks in the figures. At $kT/B < 20$, the results from the adiabatic capture scattering calculations¹³ increasingly deviate from the full and dashed lines in the figure. The results even exceed our $T \rightarrow 0$ limits.

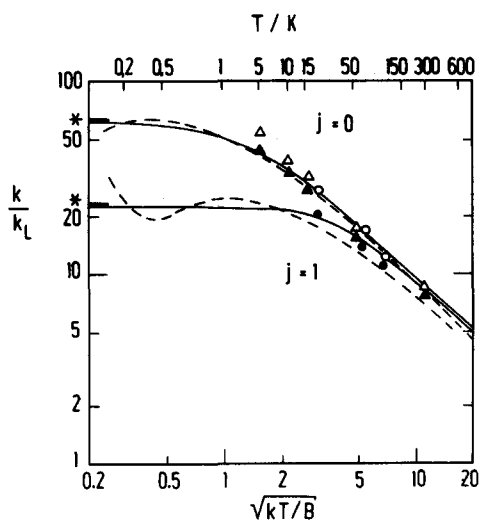


FIG. 1. State-resolved capture rate coefficients for the reaction $\text{H}_3^+ + \text{HCN}$ [$k_L = 2.27 \times 10^9 \text{ cm}^3 \text{ s}^{-1}$, $B/k = 2.13 \text{ K}$. Full lines: Eqs. (31) and (32), and $*$: $T \rightarrow 0$ limits from this work. Dashed lines: Ref. 19. Δ : $j=0$ and $\blacktriangle$: $j=1$ from Ref. 1. $\circ$: $j=0$ and $\bullet$: $j=1$ from Ref. 31, taken from Ref. 19].

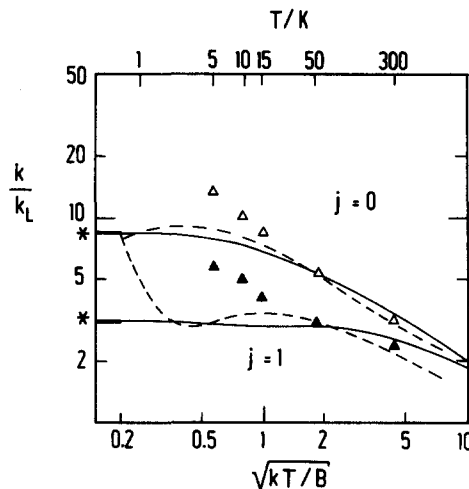


FIG. 2. State-resolved capture rate coefficients for the reaction $\text{H}_3^+ + \text{HCl}$ ($k_L = 2.28 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$, $B/k = 15.2 \text{ K}$. Symbols as in Fig. 1).

Presently, it remains unclear whether this is a numerical artifact of the ACCSA treatment or a nonstatistical effect. It cannot be due to a failure of the adiabatic assumption used in all approaches.

Figures 3 and 4 show fully thermally averaged rate coefficients for the same reactions. At $kT/B \gtrsim 20$, the present results agree well with the ACCSA results and also with the empirical representation of trajectory results from Ref. 9. At lower temperatures, the ACCSA results like in Figs. 1 and 2 exceed the present results as well as the adiabatic $T \rightarrow 0$ limit. The classical trajectory results diverge for $T \rightarrow 0$. Our calculations lead to an irregular structure at $kT/B \lesssim 5$. The origin of this effect can easily be traced back to the $j=1$ contribution. For $j=1$, the two $|m|=1$ channels are purely attractive whereas the $m=0$ channel is repulsive. At low temperatures, therefore, the threshold energy of the ($j=1, m=0$) channel cannot be overcome such that its contribution essen-

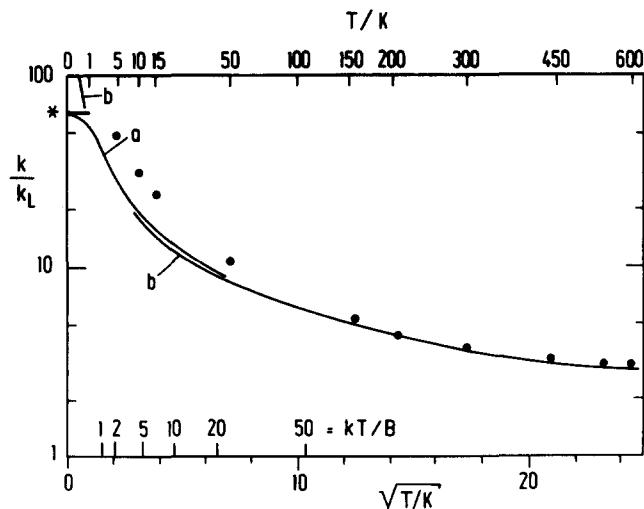


FIG. 3. Thermal capture rate coefficients $k(T)$ for the reaction $\text{H}_3^+ + \text{HCN}$ [$k_L = 2.3 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$, $B/k = 2.13 \text{ K}$. Curve a: statistical adiabatic channel results of Eqs. (31) and (32), this work; curve b: classical trajectory results, Ref. 9; $\bullet$: ACCSA results from Refs. 1 and 13].

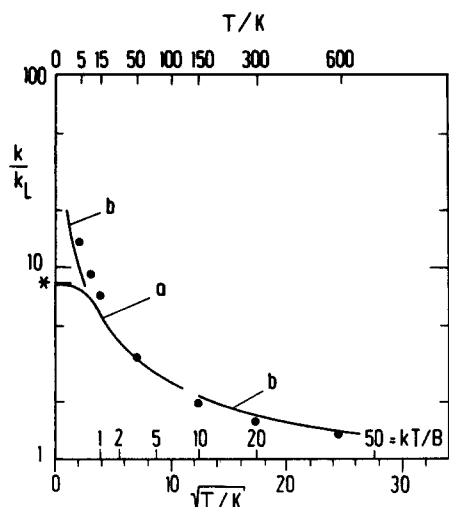


FIG. 4. Thermal capture rate coefficients $k(T)$ for the reaction $\text{H}_3^+ + \text{HCl}$ ($k_L = 2.3 \times 10^{-9} \text{ cm}^3 \text{ s}^{-1}$, $B/k = 15.2 \text{ K}$. Curves and symbols as in Fig. 3).

tially is lacking in $\bar{k}(j=1)$. The corresponding effects, for channels with $j > 1$, at higher temperatures are washed out. For high temperatures when many channels contribute, in Ref. 20 an interpolation between the threshold energies of attractive and repulsive channels (for large r and $l=0$) was proposed, i.e.,

$$E_0(j, l) \approx B j(j+1) \left\{ \frac{1 + l(l+1)/l_0(l_0+1)}{1 - l(l+1)/l_0(l_0+1)} \right\}, \quad (33)$$

with channel degeneracies $2j+1$. This grouping of m channels at low T apparently becomes insufficient. However, at high T good results are obtained. Here, dipole locking constants $C \approx 2\ln 2 - 1 = 0.386$ (defined by³² $k(T) = k_L + Ck_D$ with $k_D = 2\pi q\mu_D \sqrt{2/\pi\mu kT}$) directly follow from Eq. (33). This result is in close agreement with the trajectory results⁹ which under similar conditions give $C \approx 0.422$. The present calculations even better approach the trajectory results from Ref. 9 than our previous treatment.²⁰ Therefore, the simple expressions from Ref. 9 provide an optimum choice except at very low temperatures.

The results from Ref. 9 can be expressed by

$$\frac{k}{k_L} \approx 0.3371 y \sqrt{\frac{B}{kT}} + 0.62 \quad \text{for } \frac{kT}{B} \leq y^2/8,$$

and

$$\frac{k}{k_L} \approx 0.1900 \left(y \sqrt{\frac{B}{kT}} + 0.7198 \right)^2 + 0.9754 \quad \text{for } \frac{kT}{B} \gg y^2/8. \quad (34)$$

Figures 3 and 4 suggest that Eq. (34) is used as long as k/k_L from Eq. (34) is smaller than $k/k_L = \sqrt{1 + y^2/3}$ from Eq. (25). At lower temperatures, then Eq. (25) should be used or the more detailed result from Eqs. (31) and (32) which is also quickly calculated. For particularly simple approximate rate expressions, see the Appendix.

VI. CALCULATIONS FOR LINEAR ANISOTROPIC INDUCED DIPOLES AND QUADRUPOLES

For low temperature applications one can easily take into account the anisotropy of the polarizability α . According to Eq. (11), in the expressions from perturbation theory one just has to replace α by $\alpha_{\perp} + (\alpha_{\parallel} - \alpha_{\perp})F'(j, m)/2$. In this way, instead of Eq. (26) one obtains for $T \rightarrow 0$

$$k(j = |m|, T \rightarrow 0) = k_L^{\perp} \sqrt{1 + \alpha_A F'(j, m)/2\alpha_{\perp} + F(j, m)y_1^2} \quad (35)$$

with $k_L^{\perp} = 2\pi q \sqrt{\alpha_{\perp}/\mu}$ and $y_1 = \mu_D/\sqrt{\alpha_{\perp}B}$, which for the lowest channels is identical with Eq. (26). As long as the "attractive channels" with $F(j, m) > 0$ dominate at low temperatures, also Eq. (31) applies with k_L and y modified according to Eq. (35).

With increasing shift of the channel maxima toward smaller distances r , i.e., with increasing excitation or temperature, the perturbation results become invalid. Then, again the expansion around the torsional oscillator limit becomes applicable. Superimposed $\cos \vartheta$ and $\cos^2 \vartheta$ potentials from permanent and anisotropic induced dipole moments complicate the situation. In the following, we only consider the case of dominant anisotropic induced dipole moment. Instead of Eqs. (5), (6), (17)–(19), we now have

$$\epsilon = \frac{V_i(r) + \alpha_{\perp} q^2/2r^4 + [j(j+1) - l(l+1)]\hbar^2/2\mu r^2}{B + \hbar^2/2\mu r^2} \quad (36)$$

and

$$\lambda = \frac{(\alpha_{\parallel} - \alpha_{\perp})q^2/2r^4}{B + \hbar^2/2\mu r^2}. \quad (37)$$

Perturbation theory²⁹ for small λ gives the result

$$\epsilon \approx j(j+1) - \lambda F'(j, m)/2 \quad (38)$$

used previously for the derivation of Eq. (11). For large values of λ , one has instead²⁹

$$\epsilon \approx -\lambda + 2c\sqrt{\lambda} - d, \quad (39)$$

with

$$\begin{aligned} c &= 2\nu + |m| + 1, \\ d &= 2\nu(\nu + |m| + 1) + |m| + 1, \\ 2\nu &= j - |m| \quad \text{for } j - |m| \text{ even,} \\ 2\nu &= j - |m| - 1 \quad \text{for } j - |m| \text{ odd.} \end{aligned} \quad (40)$$

Tabulations for intermediate λ values are also available (up to $j=6, m=2$, see Ref. 29). Still assuming $B \gg \hbar^2/2\mu r^2$, this leads to channel threshold energies

$$\frac{E_{0i}}{B} \approx \left[c \sqrt{\frac{\alpha_{\parallel} - \alpha_{\perp}}{\alpha_{\perp}}} + \frac{\hbar^2}{\mu q \sqrt{8\alpha_{\parallel} B}} \right] \times [l(l+1) - j(j+1) - d]^2. \quad (41)$$

For isotropic polarizability and sufficiently large l , Eq. (41) approaches Eq. (13). By analogy with Eq. (33), we use an interpolation between Eqs. (13) and (41), which neglects the $|m|$ fluctuations in c and employs $j(j+1) - d \ll l(l+1)$, i.e.,

$$\frac{E_{oi}}{B} \approx j(j+1) + \left[j \sqrt{\frac{\alpha_{\parallel} - \alpha_{\perp}}{\alpha_{\parallel}}} + \frac{\hbar^2 l(l+1)}{\mu q \sqrt{8\alpha_{\parallel} B}} \right]^2 \quad (42)$$

Obviously Eq. (42) is most suited for high temperature application. Calculating the activated complex pseudopartition function (1) with Eqs. (22)–(24) and Eq. (42) leads to a capture rate coefficient

$$k \approx k_{\parallel} \left\{ \frac{2}{\sqrt{\pi}} \int_0^{\infty} dx \int_0^{\infty} dy \exp(-x - [K_{\alpha} \sqrt{x} + y]^2) \right\}, \quad (43)$$

which is independent of the temperature, where

$$K_{\alpha} = \sqrt{(\alpha_{\parallel} - \alpha_{\perp})/\alpha_{\parallel}}, \quad (44)$$

and $k_{\parallel}$ equals k_L with α replaced by $\alpha_{\parallel}$. The bracket equals unity for $K_{\alpha} = 0$, it decreases with increasing K_{α} , e.g., it is equal to 0.42 for $\alpha_{\perp} = 2(\alpha_{\parallel} - \alpha_{\perp})$. Equation (43) describes anisotropy reductions of the Langevin rate coefficient such as observed experimentally.⁸ With Eq. (43) they are easily expressed in terms of the parameter K_{α} . Since these effects are typical for high temperatures, we do not further investigate the quality of the approximations involved in our treatment.

We further consider collisions with a pure quadrupole in the absence of a dipole moment or major polarizability contributions. In the perturbation limit of Eqs. (11) and (12), the channel potential curves have maxima

$$E_{oi} \approx Bj(j+1) + \frac{\{l(l+1)\hbar^2\}^3}{16q^2Q^2\mu^3\{1 - 3F'(j,m)/2\}^2} \quad (45)$$

For $j = m = 0$, $E_{oi} \rightarrow \infty$ such that $k(T \rightarrow 0) \rightarrow 0$. The limiting low temperature rate coefficients for other channels directly follow from Eq. (45) via Eqs. (1) and (22) with the result

$$k(n, T \rightarrow 0) = \Gamma(4/3) \sqrt{2\pi/\mu kT} \times \{16kTq^2Q^2(1 - 3F'(j,m)/2)^2\}^{1/3}. \quad (46)$$

A further elaboration for higher temperatures, when a transition to the torsional oscillator approximation occurs, can be done quite analogous as described for the anisotropic induced dipole since $\cos^2 \vartheta$ potentials are valid for both cases. Simultaneous treatment of quadrupoles and induced dipoles appears possible only by numerical methods. This work as well as a comparison with other quadrupole calculations^{8,10,18,33,34} will not be presented here.

VII. CALCULATIONS FOR NONLINEAR DIPOLES

So far we have considered capture processes of linear molecules. The results can be generalized to symmetric and asymmetric top molecules. The Schrödinger equation (2) in a body-fixed coordinate system has to be generalized.¹² Adiabatic channel potential curves for large distances again follow from first and second order perturbation theory^{26,35}; for small distances, series expansions around the torsional oscillator limit can be made^{36–38}; for the intermediate range, extensive eigenvalue tabulations are available.³⁶ For illustrations of adiabatic channel potential curves of asymmetric rotors, see e.g., Ref. 39.

We first consider the perturbation limit for a symmetric top characterized by the quantum numbers, l, j, m, K . The $r \rightarrow \infty$ levels $Bj(j+1)$ have to be replaced by $Bj(j+1) + (A-B)K^2$. For $K=0$, the same expressions apply as for a linear molecule, i.e., Eq. (11). For $K \neq 0$, there is a contribution from first order perturbation theory; also the second order term is modified

$$V_i(r) \approx Bj(j+1) + (A-B)K^2 + \frac{l(l+1)\hbar^2}{2\mu r^2} - \frac{q\mu_D}{r^2} \left(\frac{Km}{j(j+1)} \right) + \frac{q^2\mu_D^2}{2Br^4} \times \left[\frac{(j^2 - K^2)(j^2 - m^2)}{j^3(2j-1)(2j+1)} - \frac{[(j+1)^2 - K^2][(j+1)^2 - m^2]}{(j+1)^3(2j+1)(2j+3)} \right] - \frac{q^2\alpha}{2r^4}. \quad (47)$$

In the following, the large bracket of Eq. (47) will be termed $F''(j,m,K)$. For low temperature applications the “attractive channels” (for $l=0$) dominate. For $K=0$, these are the same as for the linear molecule. For $K \neq 0$, the channels with $Km > 0$ also are attractive. However, their channel maxima are equal to the $r \rightarrow \infty$ levels as long as

$$l(l+1) \leq l_0(l_0+1)[Km/j(j+1)]. \quad (48)$$

For larger l , the $K \neq 0$ channel maxima with $Km > 0$ are equal to

$$E_{oi} \approx Bj(j+1) + (A-B)K^2 + \frac{B[G - Km/j(j+1)]^2}{2[1/y^2 - F''(j,m,K)]}. \quad (49)$$

Repulsive channels, as for linear molecules, have to be treated from the torsional oscillator side. A transition to this limit becomes also relevant for attractive channels at large l values because of the shifts of the channel maxima toward small r . In this case, instead of Eq. (17) one has³⁶

$$\epsilon \approx -\lambda + (2j - |m+K| + 1)\sqrt{2\lambda}. \quad (50)$$

The corresponding channel threshold energies by analogy with Eq. (21) follow as

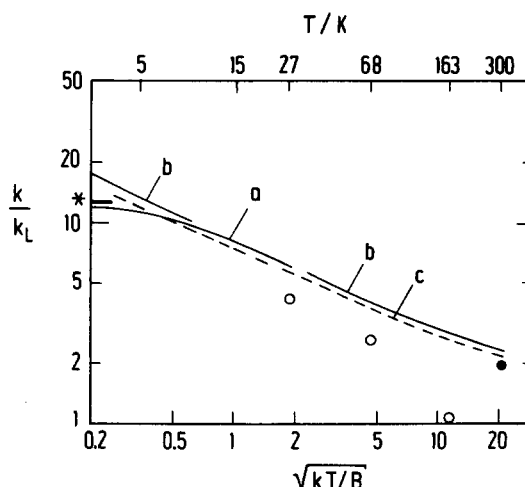


FIG. 5. Capture rate coefficients $k(T)$ for $N^+ + NH_3$ collisions (curve a: this work; curve b: Ref. 9; curve c: Ref. 12; O: CRESU experiments, ●: SIFT experiments, see Ref. 2; *: limiting rate coefficient for $T \rightarrow 0$ K, this work).

$$E_{0i} \approx B \left\{ \frac{(2j - |m + K| + 1)^2}{2(1 - G)} \right\}. \quad (51)$$

Low temperature rate coefficients are calculated from Eqs. (22)–(24). For the limit $T \rightarrow 0$ and $K = 0$, the same expressions of $k(n, T \rightarrow 0)$ are obtained as with the linear molecule, i.e., Eq. (26) for all channels with $F(j, m) > 0$. For $K \neq 0$, different results are derived. Here, $k(n, T \rightarrow 0) = 0$ for $Km < 0$. For $Km > 0$,

$$\begin{aligned} k(n, T \rightarrow 0) &\approx 2\pi q \mu_D \sqrt{2/\pi \mu k T} [Km/j(j+1)] \\ &= k_L y \sqrt{\frac{2B}{\pi k T}} [Km/j(j+1)]. \end{aligned} \quad (52)$$

Equation (52) diverges for $T \rightarrow 0$. However, in thermal averages this divergence disappears because of the Boltzmann factors of the corresponding rotor levels. For an extension toward higher temperatures, we have again compared accu-

rate evaluations of the integrals in Eqs. (27) and (28) based on tabulated eigenvalues³⁶ with approximate simple expressions such as Eqs. (31) and (32) for $K = 0$. For $K \neq 0$, the following results are derived: by analogy with Eq. (31), for $Km \geq 0$ and $F''(j, m, K) < 1/y^2$, one has

$$\begin{aligned} \frac{k(n, T)}{k_L} &\approx y \sqrt{\frac{2B}{\pi k T}} Km/j(j+1) \\ &+ \left\{ 1 - \exp \left[-I_0'' \left(\frac{1 + 1.5I_0''}{I + I_0''} \right) \right] \right\} \\ &\times \left([1 - Km/j(j+1)] y \sqrt{\frac{2B}{\pi k T}} \right. \\ &\left. + \exp \left\{ -[a(j, m, K)]^{4/3} y^{2/3} \frac{B}{kT} \right\} \right) \end{aligned} \quad (53)$$

with

$$I_0'' = \sqrt{\pi k T} [1 - y^2 F''(j, m, K)] / \{2By^2 [1 - Km/j(j+1)]^2\}.$$

By analogy with Eq. (32) for $Km \geq 0$ and $F''(j, m, K) > 1/y^2$, one derives

$$\begin{aligned} \frac{k(n, T)}{k_L} &\approx y \sqrt{\frac{2B}{\pi k T}} Km/j(j+1) + \frac{(1 - Km/j(j+1))^2}{(1 + Km/j(j+1) + a^2(j, m, K)B/2kT)} \exp \left\{ -\frac{\left(\frac{a^2(j, m, K)}{2} - j(j+1) \right) B}{(1 - Km/j(j+1))kT} \right\} \\ &\times \left[y \sqrt{\frac{2B}{\pi k T}} + \exp \left(-\frac{3}{4} [a(j, m, K)]^{4/3} y^{2/3} \frac{B}{kT} \right) \right], \end{aligned} \quad (54)$$

where $a(j, m, K) = 2j - |m + K| + 1$.

For $Km < 0$, by analogy with Eq. (32) one obtains

$$\begin{aligned} \frac{k(n, T)}{k_L} &\approx \exp \left\{ -\left[\frac{a^2(j, m, K)}{2} - j(j+1) \right] \frac{B}{kT} \right\} \\ &\times [1 + a^2(j, m, K)B/2kT]^{-1} \left(y \sqrt{\frac{2B}{\pi k T}} \right. \\ &\left. + \exp \left\{ -\frac{3}{4} [a(j, m, K)]^{4/3} y^{2/3} \frac{B}{kT} \right\} \right). \end{aligned} \quad (55)$$

Equations (53)–(55) provide good analytical representations of the detailed channel results and are easily applicable to arbitrary temperatures. In the following we demonstrate the quality of the derived expressions by comparison with detailed channel results.

VIII. EXAMPLES FOR ION-SYMMETRIC TOP DIPOLE COLLISIONS

In the following we compare our results with recent calculations by the semiclassical adiabatic invariance method from Ref. 19, and with the ACCSA method of Ref. 12, applied to the capture of NH_3 by ions. The rate coefficients $k(j=0, K=0, T)$, $k(j=1, K=0, T)$, and $k(j=1, K=1, T)$ were computed with Eqs. (31), (32), (47), (53), and (55). Table I shows the data. Apart from our approximate results, accurate channel results, obtained on the basis of tabulated eigenvalues from Ref. 36, are shown. The agreement with the analytical approximate results is always within a few percent. Similar agreement is found with the semiclassical adiabatic invariance results from Ref. 19, confirming again the equivalence of the two methods. For 300 K, the agreement with ACCSA points is only slightly less satisfactory although the same trends on the K depen-

TABLE I. State-resolved capture rate coefficients of NH_3 by ions [given values are $k(n, T)/k_L$; $\mu_D = 1.47$ D, $\alpha = 2.26 \text{ \AA}^3$, $B = 9.9 \text{ cm}^{-1}$. (a) this work, calculations with accurate channel eigenvalues from Ref. 36, (b) approximate analytical expressions from Eqs. (31), (32), (54), and (56) from this work; (c) semiclassical adiabatic invariance method of Ref. 19; (d) ACCSA calculations from Ref. 12, values taken from Fig. 5].

kT/B T/K		0.351 5			1.05 15			3.51 50			10.5 150			21.0 300			
j	K	a	b	c	a	b	c	a	b	c	a	b	c	a	b	c	d
0	0	11.6	11.6	12.6	10.1	10.2	10.5	7.7	7.9	7.6	5.6	5.7	5.3	4.4	4.5	4.2	4.3
1	0	4.8	4.5	4.8	4.9	4.4	4.8	4.7	4.4	4.6	4.1	4.1	4.0	3.5	3.6	3.4	2.7
1	1	9.3	9.1	8.3	7.0	6.8	6.5	5.5	5.4	5.3	4.4	4.5	4.3	4.0	3.8	3.6	3.2

dence of the rate coefficient are observed. Nevertheless, the differences between the values are probably within the calculational uncertainties.

Figure 5 shows thermal capture rate coefficients for $N^+ + NH_3$ collisions. At low temperatures a finite value of $k(T)$ is reached. With increasing temperature our results become identical with the values obtained from the classical trajectory calculations of Ref. 9 and ACCSA calculations of Ref. 12. The latter results at low temperatures again exceed the present values. The measurements from Ref. 2 are mostly below all calculations, although agreement is found for collisions with water.¹² The quickest calculation of the thermal rate coefficients $k(T)$ obviously is performed by using our expressions for very low temperatures up to a point where agreement with Ref. 9 is obtained. For higher temperatures, the simple expressions from Ref. 9 give a good representation of our results as well as of the ACCSA calculations.¹² At high temperatures with $k \rightarrow k_L$, anisotropies of the polarizability should be accounted for such as demonstrated in Sec. VI. Experimental values of $k < k_L$ may be interpreted in this way or by redissociation of the collision complex.¹² For particularly simple approximate rate expressions, see the Appendix.

IX. CONCLUSIONS

In the present work detailed statistical adiabatic channel calculations have led to analytical expressions for thermally averaged and state-selected thermal rate coefficients for ion-molecule capture and high pressure association reactions. With the state-resolved expressions for channel threshold energies, also detailed information on state-resolved cross sections is accessible.²¹ The agreement of the present results with data from several more recent approaches confirms that these methods are all very closely related or, apart from minor calculational differences, even identical with the statistical adiabatic channel model. The advantage of the present analytical results is their easy general applicability.

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APPENDIX: SIMPLIFIED RATE EXPRESSIONS

For quick applications to experiments, even simpler approximations than derived before may appear desirable. The low temperature limiting rate coefficient $k_a = k(T \rightarrow 0)$ from Eq. (25) is given by

$$k_a = 2\pi q \sqrt{\alpha/\mu} \sqrt{1 + \mu_D^2/3\alpha B}, \quad (A1)$$

for $\mu_D^2/3\alpha B \gg 1$, Eq. (A1) takes the form

$$k_a \approx 2\pi q \mu_D / \sqrt{3\mu B}. \quad (A2)$$

The high temperature limiting rate coefficients from Ref. 20 and from ⁹ Eq. (34) are well approximated by

$$k_b \approx 2\pi q \sqrt{\alpha/\mu} + 0.4 q \mu_D \sqrt{8\pi/\mu k T}. \quad (A3)$$

The transition from Eqs. (A1) and (A2) to Eq. (A3), to a first approximation, is performed by

$$k \approx \frac{k_a k_b}{k_a + k_b}. \quad (A4)$$

In reality, the transition is somewhat more abrupt, a switching function

$$k \approx k_b \{1 - \exp(-k_a/k_b)\} \quad (A5)$$

therefore works slightly better. However, in practice this does not appear to be relevant. Also, the high temperature approach of k_L by k_b in Eq. (A3) slightly differs from Eq. (34). However, contributions from polarizability anisotropies and other components of the potential here may contribute as well. In summary, the expressions (A1), (A3), and (A5) probably provide the simplest representation of our results for charge-molecule capture rate coefficients on a 10% accuracy level.

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